

Class 17

```
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```

```

samples <- data.frame(
  row.names = c("SRR2156848", "SRR2156849", "SRR2156850", "SRR2156851"),
  condition = c("control", "control", "treated", "treated")
)

```

```

library(tximport)

quant_dirs <- c(
  "/Users/yasoothaitham/Downloads/SRR2156848_quant/abundance.h5",
  "/Users/yasoothaitham/Downloads/SRR2156849_quant/abundance.h5",
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  "/Users/yasoothaitham/Downloads/SRR2156851_quant/abundance.h5"
)

```

```
txi <- tximport(  
  files = quant_dirs,  
  type = "kallisto",  
  txOut = TRUE  
)
```

```
1 2 3 4
```

```
names(txi)
```

```
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```

```
library(DESeq2)
```

```
Loading required package: S4Vectors
```

```
Loading required package: stats4
```

```
Loading required package: BiocGenerics
```

```
Loading required package: generics
```

```
Attaching package: 'generics'
```

```
The following objects are masked from 'package:base':
```

```
as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,  
setequal, union
```

```
Attaching package: 'BiocGenerics'
```

```
The following objects are masked from 'package:stats':
```

```
IQR, mad, sd, var, xtabs
```

The following objects are masked from 'package:base':

```
anyDuplicated, aperm, append, as.data.frame, basename, cbind,  
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,  
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,  
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,  
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,  
unsplit, which.max, which.min
```

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

```
findMatches
```

The following objects are masked from 'package:base':

```
expand.grid, I, unname
```

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

```
colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars
```

Loading required package: Biobase

Welcome to Bioconductor

```
Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.
```

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

```
rowMedians
```

The following objects are masked from 'package:matrixStats':

```
anyMissing, rowMedians
```

```
dds <- DESeqDataSetFromTximport(
  txi      = txi,
  colData  = samples,
  design   = ~ condition
)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

using counts and average transcript lengths from tximport

```
dds
```

```
class: DESeqDataSet
dim: 74 4
metadata(1): version
assays(2): counts avgTxLength
rownames(74): ENST00000539570 ENST00000576455 ... ENST00000332854
             ENST00000418193
rowData names(0):
colnames(4): SRR2156848 SRR2156849 SRR2156850 SRR2156851
colData names(1): condition
```

```
vsd <- varianceStabilizingTransformation(dds)
```

using 'avgTxLength' from assays(dds), correcting for library size

-- note: fitType='parametric', but the dispersion trend was not well captured by the function: $y = a/x + b$, and a local regression fit was automatically substituted. specify fitType='local' or 'mean' to avoid this message next time.

```
vst_mat <- assay(vsd)
```

```
nonzero <- apply(vst_mat, 1, var) > 0
vst_filt <- vst_mat[nonzero, ]
```

```
pca <- prcomp(t(vst_filt), scale = TRUE)
pca
```

Standard deviations (1, ..., p=4):

```
[1] 4.366962e+00 4.113342e+00 3.318141e+00 4.473106e-13
```

Rotation (n x k) = (47 x 4):

	PC1	PC2	PC3	PC4
ENST00000539570	-0.2058665087	0.039514602	-0.122552166	0.0716587544

ENST00000576455	-0.1665495794	0.025216305	-0.204457788	-0.1632052598
ENST00000474471	-0.1472937161	0.104636247	-0.190846361	-0.0197473313
ENST00000381700	-0.1867198474	-0.022163327	-0.172288228	0.2256209034
ENST00000420022	0.2186543654	-0.041357878	0.073395905	-0.1429568197
ENST00000553856	-0.0948853055	-0.218899071	-0.039951074	-0.0397184120
ENST00000556126	0.0269619810	-0.201448021	0.164936481	0.0005523552
ENST00000483851	-0.1913078075	-0.132992633	0.015926466	-0.0202760794
ENST00000558098	0.2227137575	-0.044684138	-0.042939418	-0.0089059901
ENST00000305831	0.0347513960	0.043685244	-0.292919035	-0.0434610257
ENST00000361390	0.2138724373	-0.005711021	0.107461070	-0.0191429209
ENST00000361453	0.1321418098	-0.063212417	-0.233325509	0.0216424576
ENST00000361624	0.2071041301	-0.083496429	-0.076286445	0.0317007331
ENST00000361739	0.1951110541	-0.068751428	0.132757926	0.0601107141
ENST00000361851	-0.1755365919	-0.104215472	0.144100045	-0.0523495498
ENST00000361899	0.2073317570	0.084557845	-0.073362837	-0.0239800626
ENST00000362079	0.2114996247	0.053424642	-0.094657794	0.0454378471
ENST00000361227	0.0180271248	0.237110038	-0.062176397	-0.0661065721
ENST00000361335	-0.1633658716	0.156989192	-0.082009672	-0.0741938029
ENST00000361381	-0.0175863484	0.237781487	0.058335936	0.0782408555
ENST00000361567	0.0482236447	0.237642304	0.003531129	-0.0334221769
ENST00000361681	0.1233694578	0.069702231	0.238741643	0.1834708738
ENST00000361789	-0.0966055073	0.205739456	0.098045667	-0.2328030276
ENST00000321039	0.0612281299	0.188386521	0.172611111	-0.2134854590
ENST00000343280	-0.1110733630	0.069634652	0.249008391	0.0677143270
ENST00000427512	0.1228556989	-0.134853106	0.191668013	-0.3550612976
ENST00000450712	0.1837153212	0.032022798	-0.175470841	-0.2140340549
ENST00000523882	0.0269619810	-0.201448021	0.164936481	-0.3210440801
ENST00000519310	0.1837153212	0.032022798	-0.175470841	-0.0216191319
ENST00000523312	-0.1867198474	-0.022163327	-0.172288228	-0.3206911979
ENST00000523439	0.1864125860	0.101514988	-0.121655822	-0.2952273127
ENST00000523138	-0.2058485530	0.039395667	-0.122663200	0.0711894632
ENST00000221735	-0.0239574548	0.191588550	0.182822589	0.0579469247
ENST00000522346	-0.1940457662	-0.125018932	0.039845999	-0.0149483420
ENST00000332030	0.1837153212	0.032022798	-0.175470841	-0.0216191319
ENST00000282292	-0.1361334361	0.146469957	0.160493465	-0.0176645774
ENST00000240719	0.0378070918	-0.223822780	0.106605810	-0.0263912804
ENST00000376233	-0.0026830829	0.230228694	-0.096738566	0.0591776854
ENST00000344222	0.0544478773	-0.188416867	-0.176452600	-0.0399643905
ENST00000325134	-0.0239574548	0.191588550	0.182822589	0.0579469247
ENST00000506609	-0.1897499165	-0.124182645	-0.069019597	-0.1455770345
ENST00000196489	-0.0006109715	-0.231659414	0.091403753	0.0197610969
ENST00000428052	0.1535529292	0.179487102	-0.021887790	0.0472921019
ENST00000536878	-0.0009221621	-0.214453024	-0.141950001	0.3153307876


```
ENST00000356724 0.0363566993 -0.200336272 0.163890597 0.0610935918
ENST00000332854 -0.2089782059 0.078150634 0.076137631 -0.3027392060
ENST00000418193 0.0039045718 0.199345713 0.172428582 0.1547162580
```

```
df <- as.data.frame(pca$x)
df$sample <- rownames(df)
df$condition <- samples$condition
df
```

	PC1	PC2	PC3	PC4	sample	condition
SRR2156848	-5.3412219	-0.5624914	-2.845355	3.874788e-13	SRR2156848	control
SRR2156849	-0.6853159	4.8623975	3.019331	3.869980e-13	SRR2156849	control
SRR2156850	0.7712620	-5.1126247	2.723940	3.874119e-13	SRR2156850	treated
SRR2156851	5.2552758	0.8127186	-2.897916	3.874107e-13	SRR2156851	treated

```
list.files(path.expand("~/Downloads"), pattern = "SRR")
```

```
[1] "SRR2156848_quant" "SRR2156849_quant" "SRR2156850_quant" "SRR2156851_quant"
```

```
samples <- data.frame(
  sample = c("SRR2156848", "SRR2156849", "SRR2156850", "SRR2156851"),
  condition = c("control", "control", "treated", "treated")
)
samples
```

	sample	condition
1	SRR2156848	control
2	SRR2156849	control
3	SRR2156850	treated
4	SRR2156851	treated

```
quant_dirs <- c(
  "~/Downloads/SRR2156848_quant/quant.sf",
  "~/Downloads/SRR2156849_quant/quant.sf",
  "~/Downloads/SRR2156850_quant/quant.sf",
  "~/Downloads/SRR2156851_quant/quant.sf"
)
quant_dirs
```

```
[1] "~/Downloads/SRR2156848_quant/quant.sf"
[2] "~/Downloads/SRR2156849_quant/quant.sf"
[3] "~/Downloads/SRR2156850_quant/quant.sf"
[4] "~/Downloads/SRR2156851_quant/quant.sf"
```

```
list.files("/Users/yasoothaitham/Downloads", pattern="SRR", full.names=TRUE)
```

```
[1] "/Users/yasoothaitham/Downloads/SRR2156848_quant"
[2] "/Users/yasoothaitham/Downloads/SRR2156849_quant"
[3] "/Users/yasoothaitham/Downloads/SRR2156850_quant"
[4] "/Users/yasoothaitham/Downloads/SRR2156851_quant"
```

```
quant_dirs <- c(
  "/Users/yasoothaitham/Downloads/SRR2156848_quant/quant.sf",
  "/Users/yasoothaitham/Downloads/SRR2156849_quant/quant.sf",
  "/Users/yasoothaitham/Downloads/SRR2156850_quant/quant.sf",
  "/Users/yasoothaitham/Downloads/SRR2156851_quant/quant.sf"
)

file.exists(quant_dirs)
```

```
[1] FALSE FALSE FALSE FALSE
```

```
library(DESeq2)

samples <- data.frame(
  row.names = c("SRR2156848", "SRR2156849", "SRR2156850", "SRR2156851"),
  condition = c("control", "control", "treated", "treated")
)

dds <- DESeqDataSetFromTximport(
  txi      = txi,
  colData  = samples,
  design   = ~ condition
)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

using counts and average transcript lengths from tximport

```
dds
```

```
class: DESeqDataSet
dim: 74 4
metadata(1): version
assays(2): counts avgTxLength
rownames(74): ENST00000539570 ENST00000576455 ... ENST00000332854
             ENST00000418193
rowData names(0):
colnames(4): SRR2156848 SRR2156849 SRR2156850 SRR2156851
colData names(1): condition
```

```
df <- as.data.frame(pca$x)
df$sample <- rownames(df)
df$condition <- c("control","control","treated","treated")

head(df)
```

	PC1	PC2	PC3	PC4	sample	condition
SRR2156848	-5.3412219	-0.5624914	-2.845355	3.874788e-13	SRR2156848	control
SRR2156849	-0.6853159	4.8623975	3.019331	3.869980e-13	SRR2156849	control
SRR2156850	0.7712620	-5.1126247	2.723940	3.874119e-13	SRR2156850	treated
SRR2156851	5.2552758	0.8127186	-2.897916	3.874107e-13	SRR2156851	treated

```
setwd("~/Downloads")
```

```
setwd("~/Downloads")
```

Core Unix commands

Most unix commands have super short names, which makes them quick to type but annoying to learn. Major file system related commands include:

pwd: Print working directory ls: list files and directories cd: Change Directory mkdir: Make a new directory rm: remove files and directory (delete) : Copy files or dirs (source > destination) mv: Move files or dirs (basically re-name) nano: A wee text editor (very basic but always available)

curl: Download files from web or ftp wet: Download files from the web tar -zxvf: UnTar (un-package) Tar archive files gunzip: UnZip files \$PATH: The places (dirs) to look for programs

AWS EC2 Instance

```
ssh -i ~/Downloads/bimm143__ya.pem ubuntu@ec2-34-222-197-175.us-west-2.compute.amazonaws.com
ssh -i "bioinf_yasmeen.pem" ubuntu@ec2-34-222-197-175.us-west-2.compute.amazonaws.com
ssh -i ~/Downloads/bioinf_yasmeen.pem ubuntu@ec2-34-222-197-175.us-west-2.compute.amazonaws.com
```

Secure Copy files between machines, in this case from our instance to our laptop `scp -i ~/Downloads/bioinf_yasmeen.pem ubuntu@ec2-16-145-48-135.us-west-2.compute.amazonaws.com:~/results.tsv .`

Class 17 Instance

```
ssh -i ~/Downloads/bioinf_yasmeen.pem ubuntu@ec2-34-222-197-175.us-west-2.compute.amazonaws.com
export KEY=~/Downloads/bioinf_yasmeen.pem
export SERVER=ubuntu@ec2-34-210-54-74.us-west-2.compute.amazonaws.com
ssh -i $KEY $SERVER scp -r -i $KEY $SERVER:~/*_quant .cd ~/Downloads
```

Q1: What command prints the first lines of a FASTQ file?

```
head SRR2156848_1.fastq
```

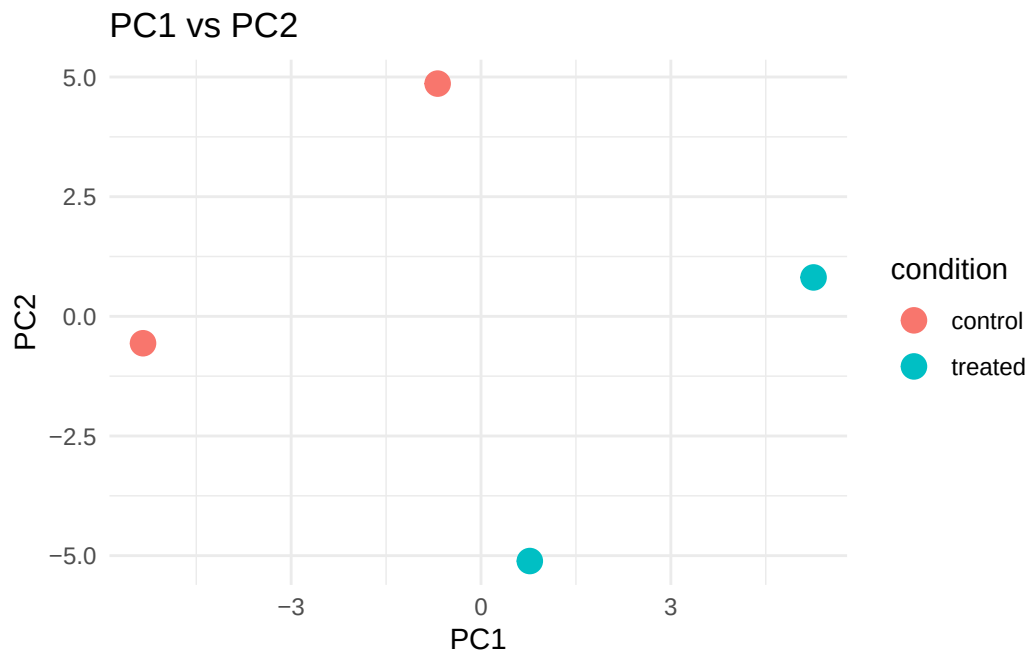
Q2: How to count the number of reads?

```
grep -c "^@" SRR2156848_1.fastqhead SRR2156848_1.fastq
```

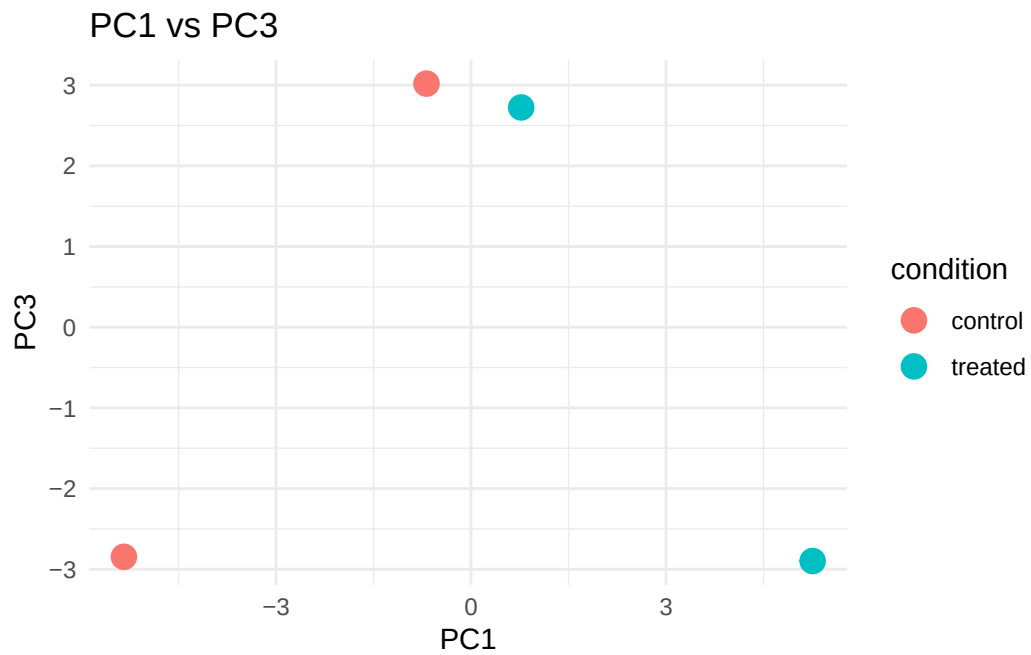
```
library(ggplot2)

df <- as.data.frame(pca$x)
df$sample <- rownames(df)
df$condition <- c("control", "control", "treated", "treated")
```

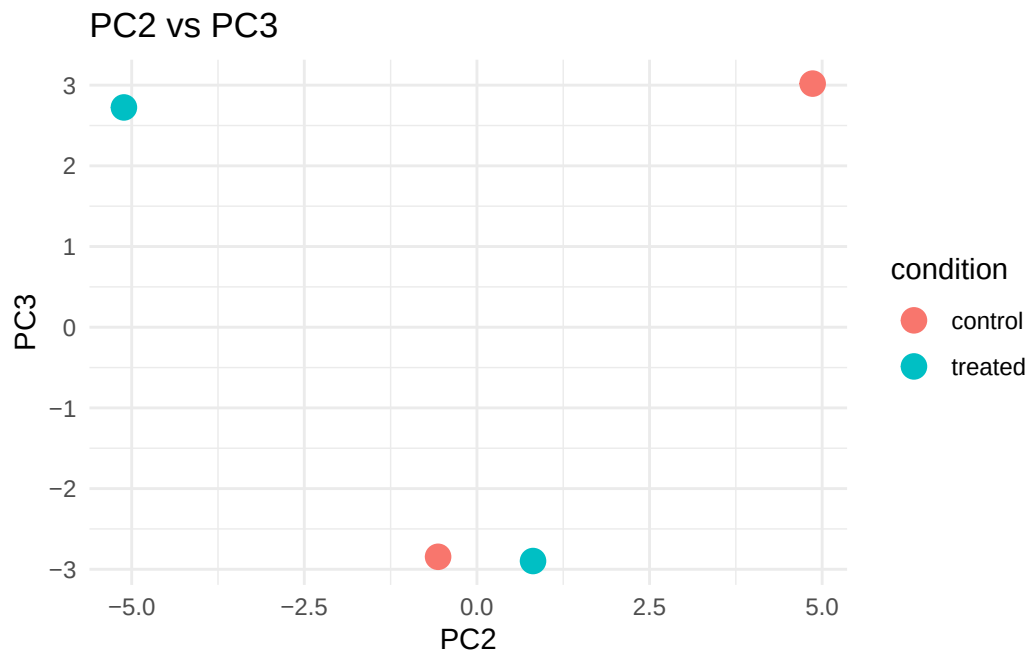
```
ggplot(df, aes(PC1, PC2, color=condition)) +
  geom_point(size=4) +
  theme_minimal() +
  labs(title="PC1 vs PC2")
```



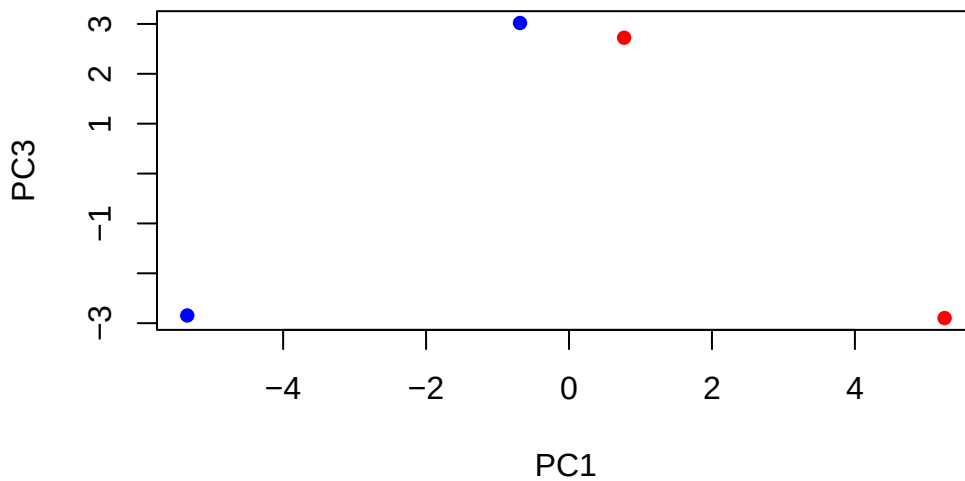
```
ggplot(df, aes(PC1, PC3, color=condition)) +  
  geom_point(size=4) +  
  theme_minimal() +  
  labs(title="PC1 vs PC3")
```



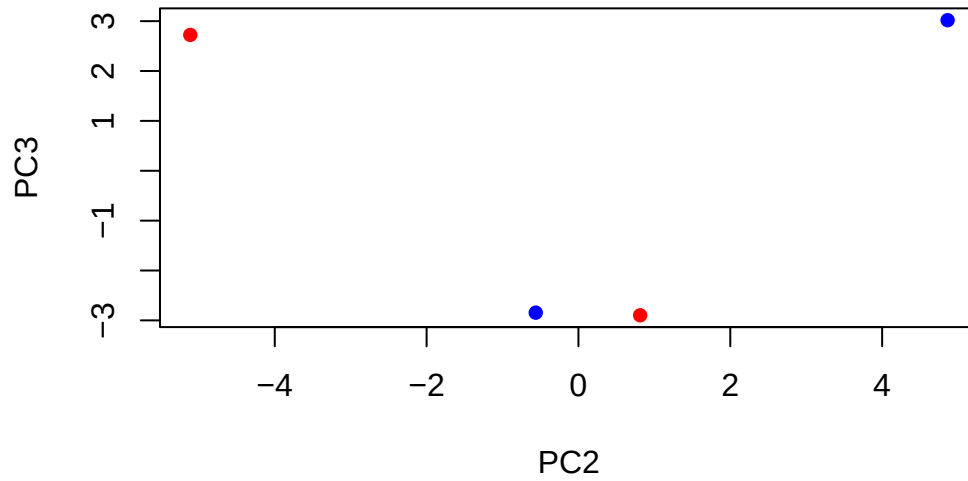
```
ggplot(df, aes(PC2, PC3, color=condition)) +  
  geom_point(size=4) +  
  theme_minimal() +  
  labs(title="PC2 vs PC3")
```



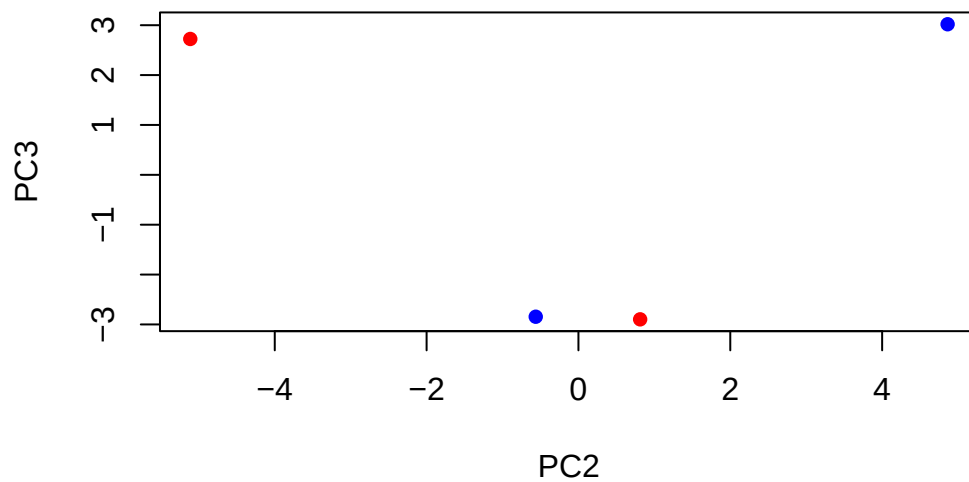
```
plot(pca$x[,1], pca$x[,3],  
     col=c("blue", "blue", "red", "red"),  
     xlab="PC1", ylab="PC3", pch=16)
```



```
plot(pca$x[,2], pca$x[,3],  
     col=c("blue", "blue", "red", "red"),  
     xlab="PC2", ylab="PC3", pch=16)
```



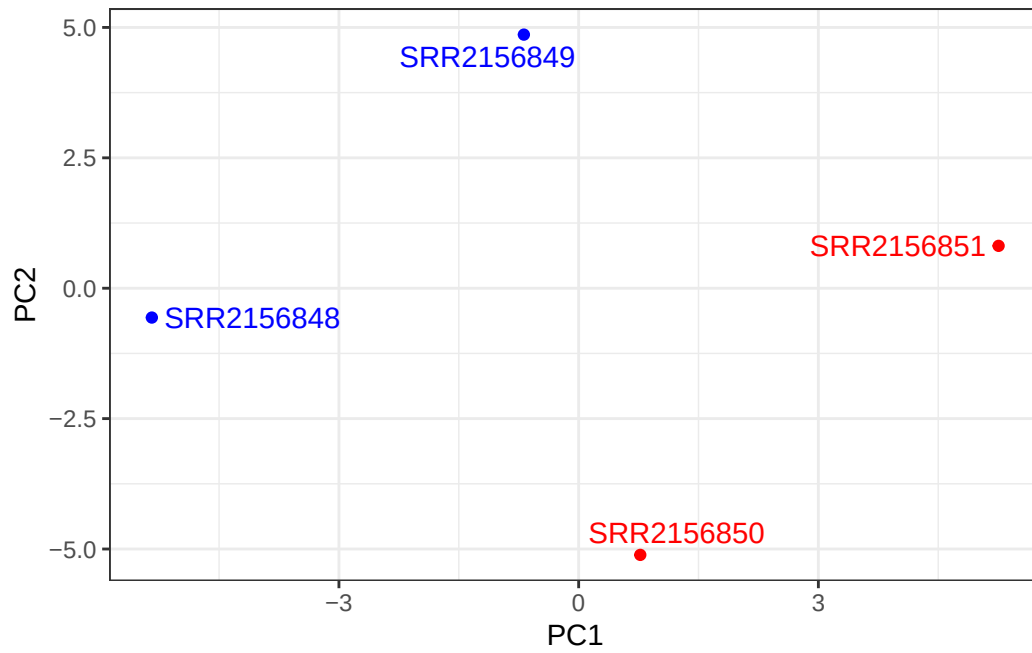
```
plot(pca$x[,2], pca$x[,3],  
     col=c("blue", "blue", "red", "red"),  
     xlab="PC2", ylab="PC3", pch=16)
```

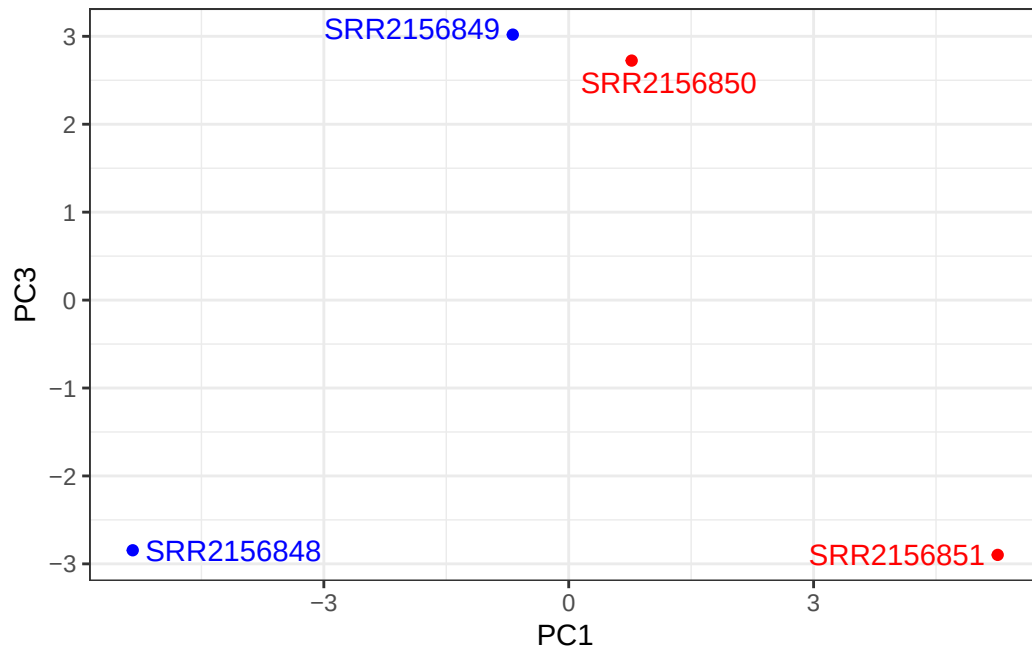
```
library(ggplot2)
library(ggrepel)

mycols <- c("blue","blue","red","red")

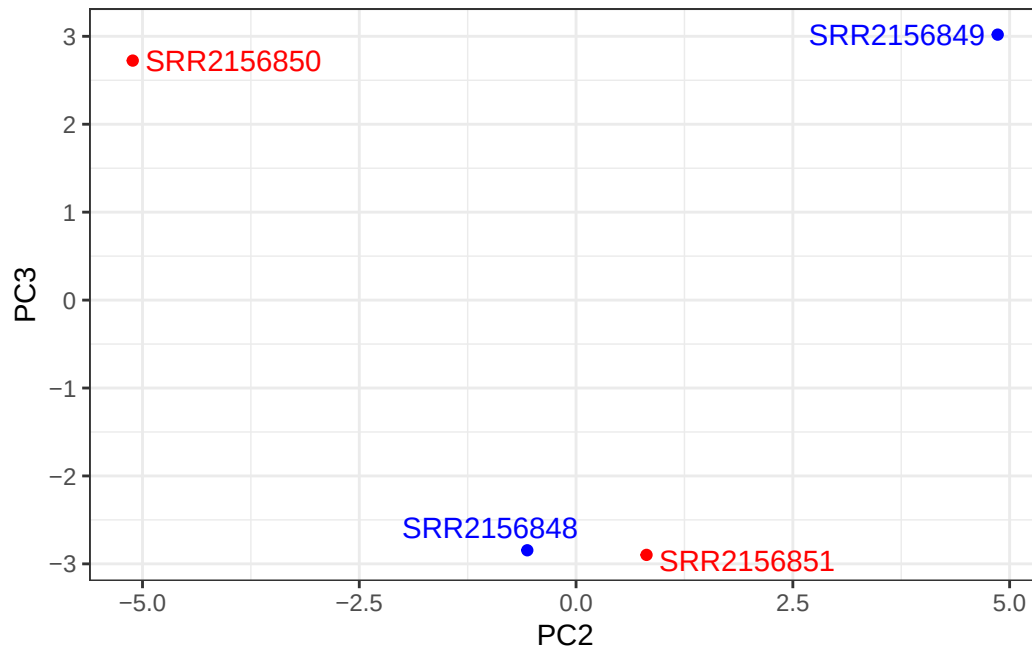
ggplot(pca$x) +
  aes(PC1, PC2, label=rownames(pca$x)) +
  geom_point(col = mycols) +
  geom_text_repel(col = mycols) +
  theme_bw()
```



```
ggplot(pca$x) +  
  aes(PC1, PC3, label=rownames(pca$x)) +  
  geom_point(col = mycols) +  
  geom_text_repel(col = mycols) +  
  theme_bw()
```



```
ggplot(pca$x) +  
  aes(PC2, PC3, label=rownames(pca$x)) +  
  geom_point(col = mycols) +  
  geom_text_repel(col = mycols) +  
  theme_bw()
```



NOTE about graphs: lab screenshot used the full Ensembl transcriptome where i had to use the MINI transcriptome. However, the good news is that it does NOT the PCA pattern! So the pattern should still be correct.